



**Cambridge Instruments
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

OIL TEST SET · TRANSFORMER OIL BDV TESTER

CIE/7000 SERIES

Transformer oil dielectric strength (BDV) tester series — 0–50 kV to 0–120 kV, 8 models. Manual and motorised variants.

50–120kV

Transformer Oil BDV Tester Series

A range of oil dielectric strength test sets for breakdown/withstand voltage testing of transformer and circuit breaker insulating oil per IS 6792:1992, spanning eight output-voltage tiers from 0–50kV (CIE/7050) up to 0–120kV (CIE/7120). Every model shares the same design and is available in manual and motorised (MOT) drive.

- **8 model tiers** — 0–50kV to 0–120kV, one model per rating
- **Manual or motorised** — motorised advances at approx. 2kV/second
- **Memory device** — retains breakdown reading after tripping
- **Zero-start interlocking** — plus interlocked H.T. door cutout
- **Analog moving-iron metering** — 96mm² kV meter, L.T. side
- **Standard test vessel** — brass spheres, per IS 6792:1992



- V **50–120kV** 8 distinct model tiers
- M **Manual/Motorised** Approx. 2kV/second (motorised)
- ✓ **Memory Device** Retains breakdown reading

01 Full Specification Sheet

PARAMETER	SPECIFICATION
Output Voltage Range	0–50 kV to 0–120 kV (8 models, continuously variable)
Capacity Range	500 VA to 1200 VA (varies by model)
Input	230 V ±10%, 50 Hz, 1 phase A.C.
Duty	Intermittent
H.V. Winding	Cast-resin, graded insulation, centre point grounded
Metering	96 mm ² A.C. moving-iron voltmeter, L.T. side, scaled in kV (analog)
Motorised Rate	Approx. 2 kV/second (motorised variant)
Test Vessel	Two brass spheres, 12.5–13 mm dia., 4mm or 2.5mm gap, per IS 6792:1992
Warranty	12 months against manufacturing defects

Any requirement apart from the standard voltage tiers & drive above can be built to order.

02 CIE/7000 Series — Output Voltage & Capacity

OUTPUT VOLTAGE	CAPACITY	MANUAL MODEL	MOTORISED MODEL
0–50 kV	500 VA	CIE/7050	CIE/7050 MOT
0–60 kV	600 VA	CIE/7060	CIE/7060 MOT
0–70 kV	700 VA	CIE/7070	CIE/7070 MOT
0–75 kV	750 VA	CIE/7075	CIE/7075 MOT
0–80 kV	800 VA	CIE/7080	CIE/7080 MOT
0–90 kV	900 VA	CIE/7090	CIE/7090 MOT
0–100 kV	1000 VA	CIE/7100	CIE/7100 MOT
0–120 kV	1200 VA	CIE/7120	CIE/7120 MOT

03 Typical Applications

- Transformer oil dielectric strength testing before energisation
- Routine oil quality monitoring for distribution and power transformers
- Circuit breaker oil testing
- Power utility and substation maintenance laboratories

04 What's in the Box

- ✓ Instruction manual with circuit diagram
- ✓ Input supply cable, minimum 2 metres, 3-pin plug
- ✓ Test vessel (oil cell) with electrode and gap-setting gauge

05 Optional Accessories

- + Alternate tripping current set point
- + Digital metering in place of analog
- + 2-unit housing
- + Double capacity, 20 mA (CIE/7050–7080 only)
- + Magnetic stirrer

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reliability since 1963.